Point-slope form

1 State what form is the equation $y = 3x + 6$ written in.

2 Write down the point-slope form of the equation of a non-vertical line with slope m that passes through the point (x_1, y_1).

3 Given a straight line with slope 1 going through point $A\,(2, 4)$, is the slope between A and any other point on the line always 1?

4 For each of the following:

- **i** By substituting into the equation $y = mx + b$, find the value of b for this line.
- **ii** Hence, write down the equation of the line in slope-intercept form.
- **iii** Find the x-intercept of the line.
- **iv** Sketch the graph of the line.

a A line has slope $\frac{2}{3}$ and passes through the point $(-6, 2)$.

b A line has slope -4 and passes through the point $(3, -8)$.

5 A line has slope 2 and passes through the point $(-1, 5)$.

a By substituting into the equation $y = mx + b$, find the value of b for this line.

b Hence, write down the equation of the line in slope-intercept form.

c Use the point-slope formula to arrive at the same equation from part (b). Show your work.

d Sketch the graph of the line.

6 A line has a slope of -10 and intersects the y-axis at -1.

a Use the point-slope formula to find the equation of the line in standard form.

b Does the point $(-7, 68)$ lie on this line?

7 Use the point-slope formula to find the equation of the following lines in standard form.

a A line passes through the point $(3, -5)$, and has a slope of $-\frac{9}{2}$.

b A line has a slope of $\frac{5}{9}$ and intersects the x-axis at -10.

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8 $A(-1,4)$ is a point on a straight line.
If the slope of the line is 4, determine whether the following statements are correct.

a $\dfrac{x+1}{y-4}=4$ **b** $\dfrac{y-4}{x+1}=4$ **c** $\dfrac{y+1}{x-4}=4$ **d** $\dfrac{4-y}{-1-x}=4$

9 Each of the following pairs of points define a line:

i Find the slope of the line.

ii Find the equation of the line in slope-intercept form.

a $(0,5)$ and $(5,25)$

b $(0,-9)$ and $(3,12)$

c $(0,3)$ and $(-7,31)$

d $(0,8)$ and $(8,80)$

e $(0,-6)$ and $(-10,-56)$

f $(6,3)$ and $(9,7)$

g $(1,3)$ and $(-3,-13)$

h $(2,-7)$ and $(-5,6)$

10 A line passes through the points $A(3,4)$ and $B(-3,-9)$.

a Find the slope of the line.

b Find the equation of the line using the point-slope formula, and any one point. Express your answer in the form $Ax+By=C$ with $A>0$.

11 Using the point-slope formula, find the equation of the following lines in slope-intercept form.

a A line passes through $(-2,-9)$ and has a slope of -2.

b A line passes through $\left(\dfrac{7}{8},-5\right)$ and has a slope of 2.

c A line passes through $(7,6)$, and has a slope of $\dfrac{9}{2}$.

d A line passes through $(-4,-2)$, and has a slope of 3.

e A line passes through $(-6,1)$ and has a slope of 4.

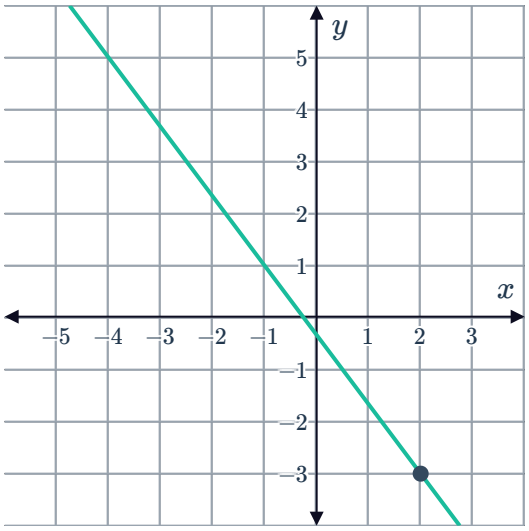
f A line passes through $\left(1,-\dfrac{3}{4}\right)$ and has a slope of 3.

g A line passes through $(-5,9)$ and has a slope of $\dfrac{3}{2}$.

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Additional questions

12 State the equation of the given linear function in point-slope form.



13 Sketch the graph of the lines on the coordinate plane given the following equations:

a $y + 8 = 3(x + 2)$

b $y + 1 = -2(x + 1)$

c $y + 6 = -\frac{1}{3}(x - 3)$

d $y + 2 = \frac{1}{4}x$

14 Sally receives an order to make key chains for an upcoming festival. She is told to make at least 50 and at most 100 key chains, and will be paid $\$8$ for each key chain she makes. Sally makes 80 key chains and is paid $\$665$ altogether.

a Let x represent the number of key chains Sally makes and y represent the amount she is paid. Write an equation in point-slope form relating x and y.

b Convert the equation from part (a) to slope-interept form.

c Determine the flat fee that Sally was paid in addition to the $\$8$ per key chain that she received.

d State the domain of this linear relationship

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15 Gary is flying in a plane which is moving directly away from his home at a constant speed. At 2 hours into the flight, he checked that he was 1042 miles from his home. He checks again at 7 hours into the flight to find that he is now 3542 miles from his home.

- **a** Find the speed of the plane, in miles per hour.
- **b** Write an equation in point-slope form relating the number of hours flying, x, to the distance Gary is from his home, y, in miles.
- **c** If the destination of Gary's flight is 5542 miles from his home, find its duration.

16 Create a scenario where you would choose to use each of the following forms and explain why you would use that form over the others:

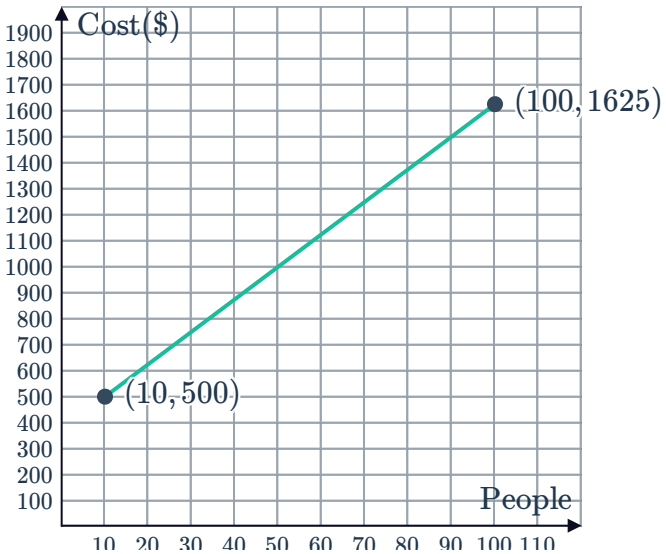
- **a** Point-slope form, $(y - y_1) = m(x - x_1)$
- **b** Slope-intercept form, $y = mx + b$
- **c** Standard form, $Ax + By = C$

17 Rewrite the point-slope form of a linear equation in slope-intercept form. Draw connections between the two forms.

18 A cup is placed under a faucet which is leaking at a rate of 2 oz per hour. After 4 hours, the cup contains 15 oz of water.

- **a** Write an equation in point-slope form relating the volume in the cup, y, to the amount of time that has passed in hours, x.
- **b** Now, rewrite the equation into slope-intercept form.
- **c** State and interpret the y-intercept of the linear equation.
- **d** Suppose that the cup has a maximum capacity of 18 oz. Determine how long it can be under the faucet before it starts to over flow. Now, state the domain and range.

19 Paige is planning an event and the brochure for the caterer at the venue shows the following graph for the meal costs.



- **a** State and interpret the domain of the linear function.
- **b** Paige concludes that the unit cost per person is $\$16.25$. Determine if this is correct. If so, explain how to calculate this value. If not, explain the error and state the correct value.
- **c** State the equation of the linear function in point-slope form

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